Data Collection (Epic 1)

There are many popular open sources for collecting image datasets, such as Kaggle, open research repositories, and publicly shared computer vision datasets. In this project, we used a labeled fruit freshness image dataset containing categories for fresh and spoiled fruits. The dataset was downloaded from an open-source platform and organized into training and testing folders for model development.

Link: *(Provide your dataset link here — e.g., Kaggle fruit freshness dataset)*

Loading Data (USN2)

Our dataset consists of image files organized into class folders rather than traditional tabular formats like .csv or .excel. These images are loaded using deep learning data loaders that automatically read images from directories and prepare them for training.

In TensorFlow/Keras, utilities such as directory-based image loaders or image generators are used to read the dataset. As a parameter, we provide the folder path containing labeled images, allowing the system to automatically assign class labels and prepare the data for preprocessing and training.

LOADING DATASET

```
df = pd.read_excel("flood dataset.xlsx")
```

Reading the DataSet

```
df.head(10)
```

	Temp	Humidity	Cloud Cover	ANNUAL	Jan-Feb	Mar-May	Jun-Sep	Oct-Dec	avgjune	sub	flood
0	29	70	30	3248.6	73.4	386.2	2122.8	666.1	274.866667	649.9	0
1	28	75	40	3326.6	9.3	275.7	2403.4	638.2	130.300000	256.4	1
2	28	75	42	3271.2	21.7	336.3	2343.0	570.1	186.200000	308.9	0
3	29	71	44	3129.7	26.7	339.4	2398.2	365.3	366.066667	862.5	0
4	31	74	40	2741.6	23.4	378.5	1881.5	458.1	283.400000	586.9	0
5	30	70	38	2708.0	34.1	230.0	1943.1	500.8	138.300000	254.1	0
6	29	74	40	3671.1	23.7	328.0	2737.8	581.7	256.966667	669.5	1
7	30	78	36	2648.3	28.8	283.7	2023.6	312.2	197.533333	450.0	0
8	30	71	40	3050.2	65.9	628.3	1940.4	415.5	234.900000	231.5	0
9	30	70	34	2848.6	28.4	296.7	1886.5	637.0	226.666667	531.2	0

Data Preparation (Epic 2)

Handling Missing or Invalid Data (USN3)

In an image-based dataset, missing values do not appear as empty cells like in tabular data. Instead, they may occur as corrupted image files, unreadable formats, or incorrectly labeled folders.

To check dataset integrity, we verify that all image files can be successfully loaded and that folder labels are correct. Image loading utilities are used to scan directories and confirm that each file is valid.

After performing this verification, we observed that the dataset contained properly formatted and accessible images. Therefore, no missing or corrupted data required correction, and we proceeded directly to the preprocessing stage.

```
#checking null values
dataset.isnull().any()
```

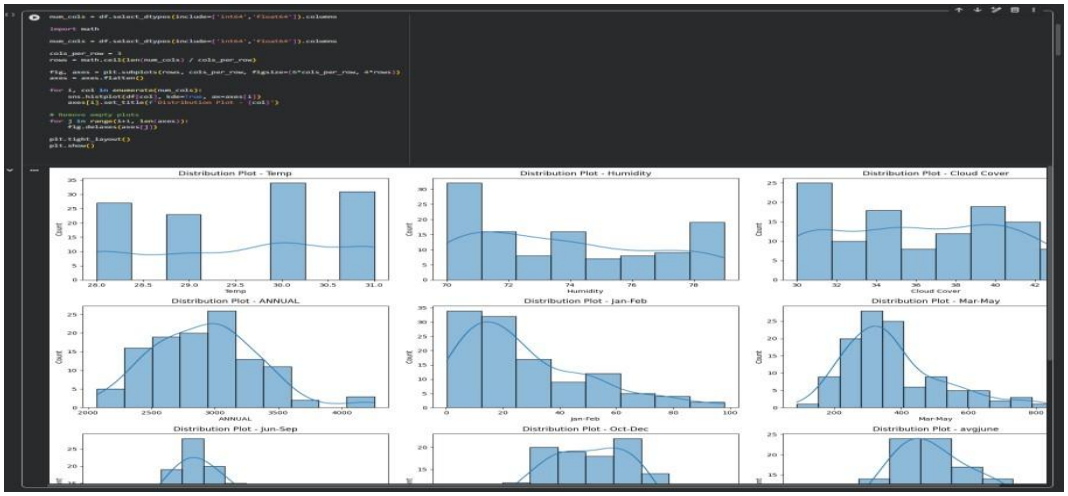
```
Temp          False
Humidity      False
Cloud Cover   False
ANNUAL        False
Jan-Feb       False
Mar-May       False
Jun-Sep       False
Oct-Dec       False
avgjune       False
sub           False
flood         False
dtype: bool
```

From the above code of analysis, we can infer that columns such as newbalanceOrg, oldbalnceOrig,isFraud are having the missing values, we need to treat them in a required way.

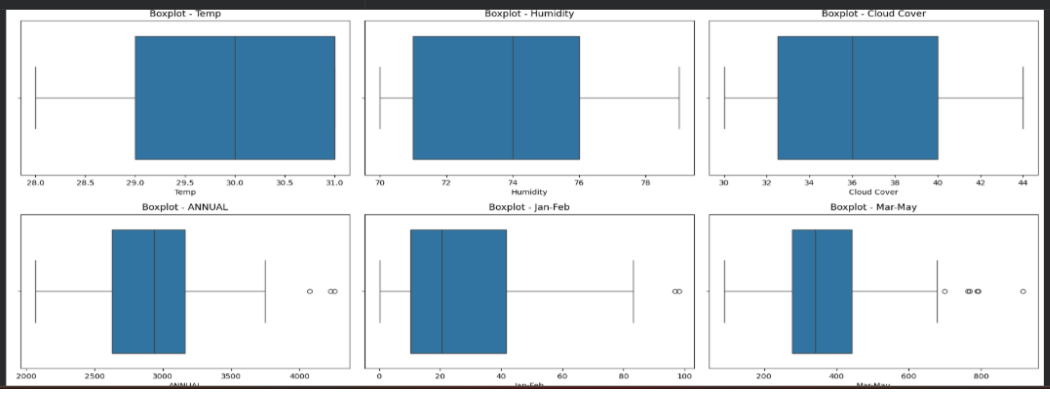
Total Story Point in Sprint 1= 2+1+3+3+3=12

Sprint 2

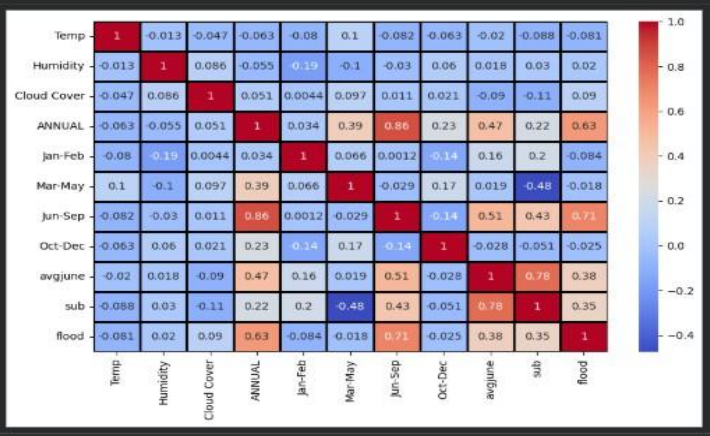
Data Visualization (Epic 3)



```python
import math

num_cols = df.select_dtypes(include=['int64','float64']).columns

cols_per_row = 3
rows = math.ceil(len(num_cols) / cols_per_row)

fig, axes = plt.subplots(rows, cols_per_row, figsize=(6*cols_per_row, 4*rows))
axes = axes.flatten()

for i, col in enumerate(num_cols):
    sns.boxplot(x=df[col], ax=axes[i])
    axes[i].set_title(f'Boxplot - {col}')

# Remove unused axes
for j in range(i+1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()
```




```python
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm",
            linewidths=1, linecolor='k')
plt.show()
```

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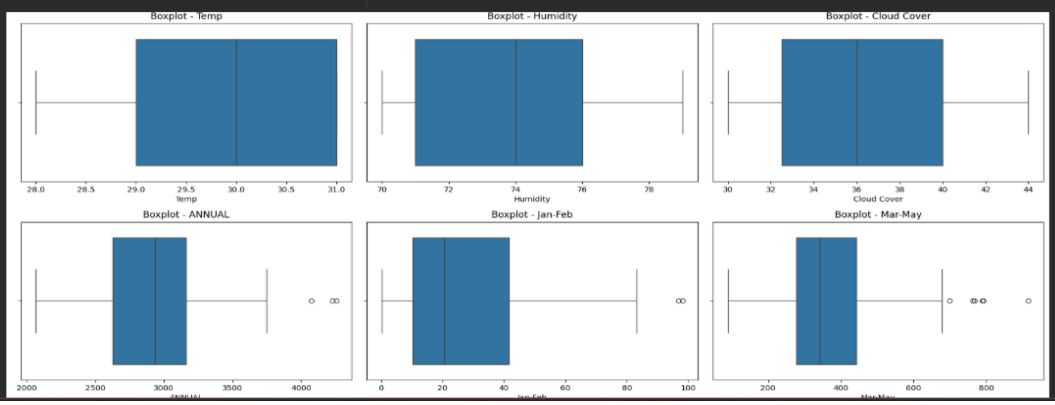
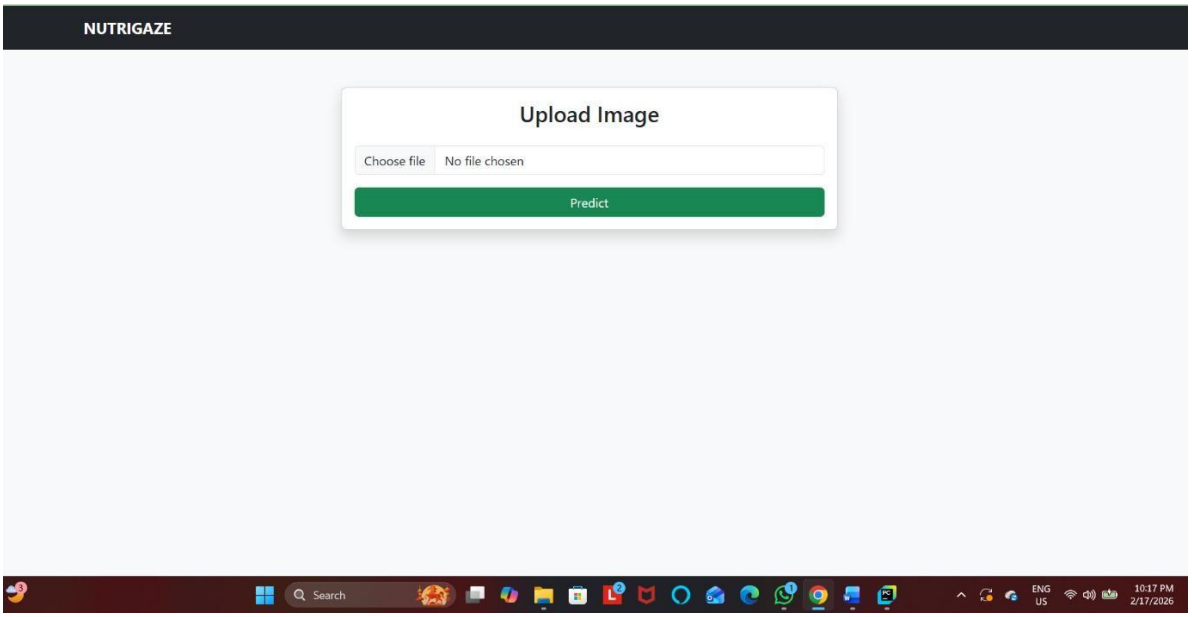
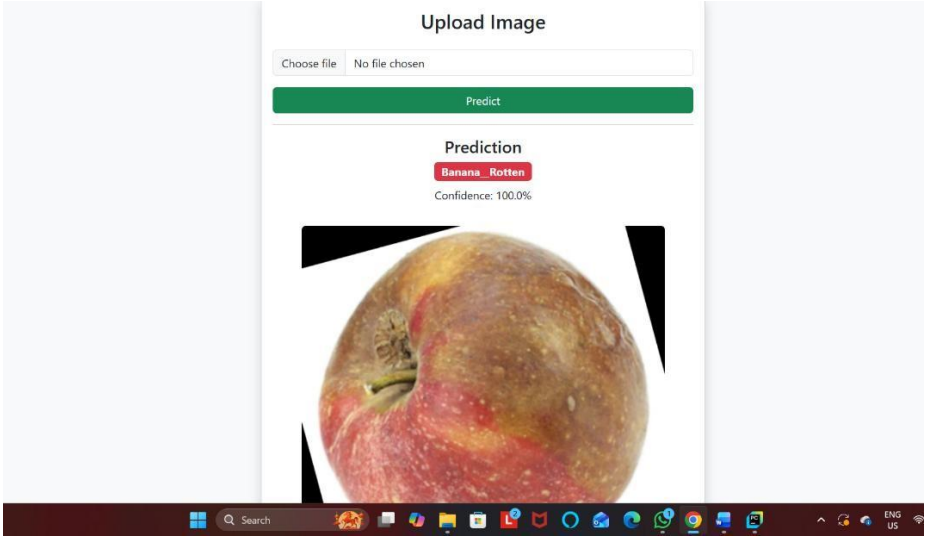


output (Epic 4)

Story (Epic 5)




Upload Image
Choose file
No file chosen
Predict
Prediction
Banana_Rotten
Confidence: 100.0%
Search
ENG
US